Product List Management System

Using Doubly and Circular Linked Lists

1. Problem Statement

To develop a Product List Management System using Doubly Linked List and Circular Linked List to store and manage product details such as product ID, product name, and price. The system supports insertion, deletion, searching, and displaying of product records.

2. Objectives

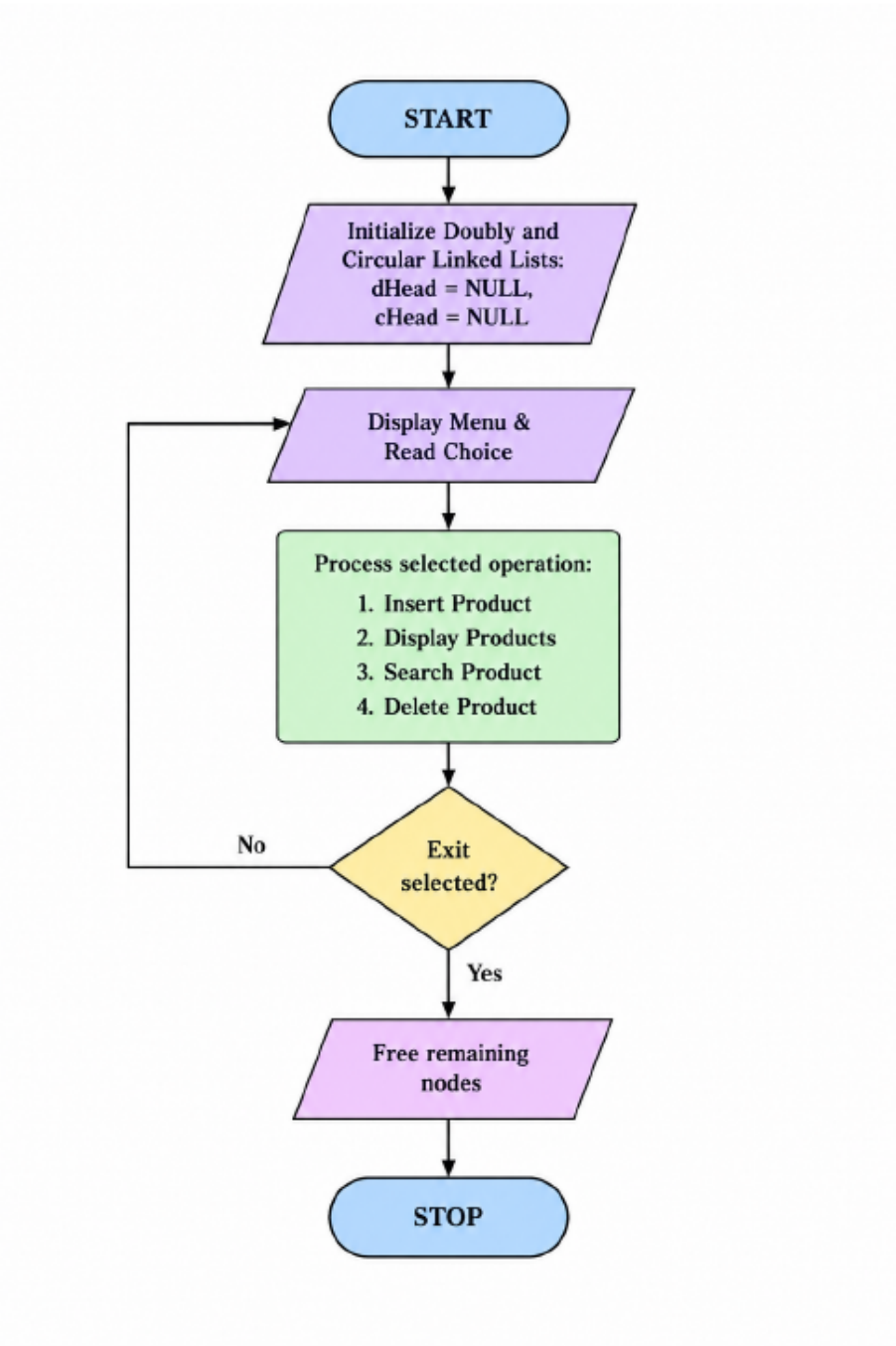
1. To store product details such as product ID, name, and price using linked lists.
2. To implement Doubly Linked List and Circular Linked List.
3. To perform insertion and deletion of product records.
4. To search for products using their ID.
5. To display all product details efficiently.

3. Algorithm

1. Start.
2. Create a product node containing ID, name, price, next pointer, and previous pointer.
3. Initialize both linked lists to NULL.
4. Display the menu of operations.
5. Insert product details into both linked lists.
6. Display products from the selected linked list.
7. Search for a product using its ID.
8. Delete a product from both linked lists using its ID.
9. Repeat the operations until the user selects Exit.
10. Stop.

4. Flowchart

The flowchart shows the main steps of the Product List Management System.



5. Conclusion

The Product List Management System demonstrates the use of Doubly and Circular Linked Lists to manage product records. It supports insertion, deletion, searching, and displaying of products and helps explain the practical applications of linked lists and dynamic memory management.